

A local AI workflow for traceable Cs-137 groundwater screening — with cited evidence, uncertainty and zero LLM-generated physics.

AMD ROCm

PHYSICS-FIRST AI

Stefano Rigante · Nuclear engineer



NuclidePath

physics-first private agent

LOCAL ONLY

AMD ROCm

SCENARIO CONTROLS

Cs-137 groundwater screening

Edit declared inputs. Every number is recomputed by the deterministic tool.

Assessment distance

100

metres

Reference concentration C_0

1000000

Bq/m³

Distribution coefficient Kd

0.2

m²/kg · literature screening value

Potassium K+

20

mg/L · competition driver

Pore-water velocity

0.00001

m/s

Porosity

0.35

fraction

Run private workflow

→

No external API calls.

LLM planning cannot alter physics outputs or issue emergency actions.

TRACK 2 · LOCALIZED AI AGENTS

Traceable screening, from evidence to uncertainty

Site characterization, local retrieval, allow-listed planning, deterministic transport and reproducible uncertainty — executed entirely on the AMD system.

VERIFIED

Effective Kd	Retardation factor	Estimated travel time	Final concentration
0.1667	810.5	256.8	3,615
m ² /kg	dimensionless	years	Bq/m ³

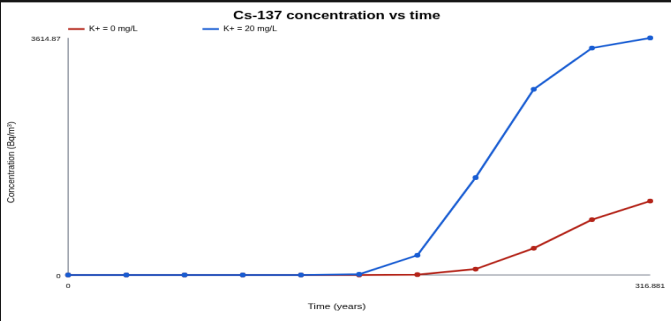
Overview

Uncertainty

Provenance

Artifacts

Concentration over time · K+ comparison



Cs-137 concentration vs time

— K+ = 0 mg/L — K+ = 20 mg/L

Concentration (Bq/m³)

Time (years)

Verified agent workflow

01 validate permissions ✓

02 site agent ✓

03 knowledge retrieval ✓

04 environment agent ✓

05 store memory ✓

06 synthesize report ✓

Track 2 capabilities

✓ local knowledge retrieval

✓ local multi turn memory

✓ multi step planning

✓ permission privacy control

✓ tool invocation

Warnings & scientific limitations

Site characterization is incomplete or non-site-specific; results are screening-only

Site and event metadata are explicitly demonstrative

Prototype model; not validated for operational radiological assessment

5/5

Track 2 capabilities

231

AMD core snapshot

376/15

latest PR gate

Technical AI fails when language becomes calculation

NuclidePath makes the responsibility boundary explicit and testable.

01 · The failure mode

A general-purpose LLM can invent parameters, equations, units or reassuring conclusions — all unacceptable in emergency-analysis workflows.

02 · The boundary

The model returns only a six-step allow-listed plan. Strict schemas feed deterministic physics, retrieval and uncertainty tools.

03 · The outcome

Every value is versioned, reproducible and tied to inputs, assumptions, sources, warnings and SHA-256 artifacts.

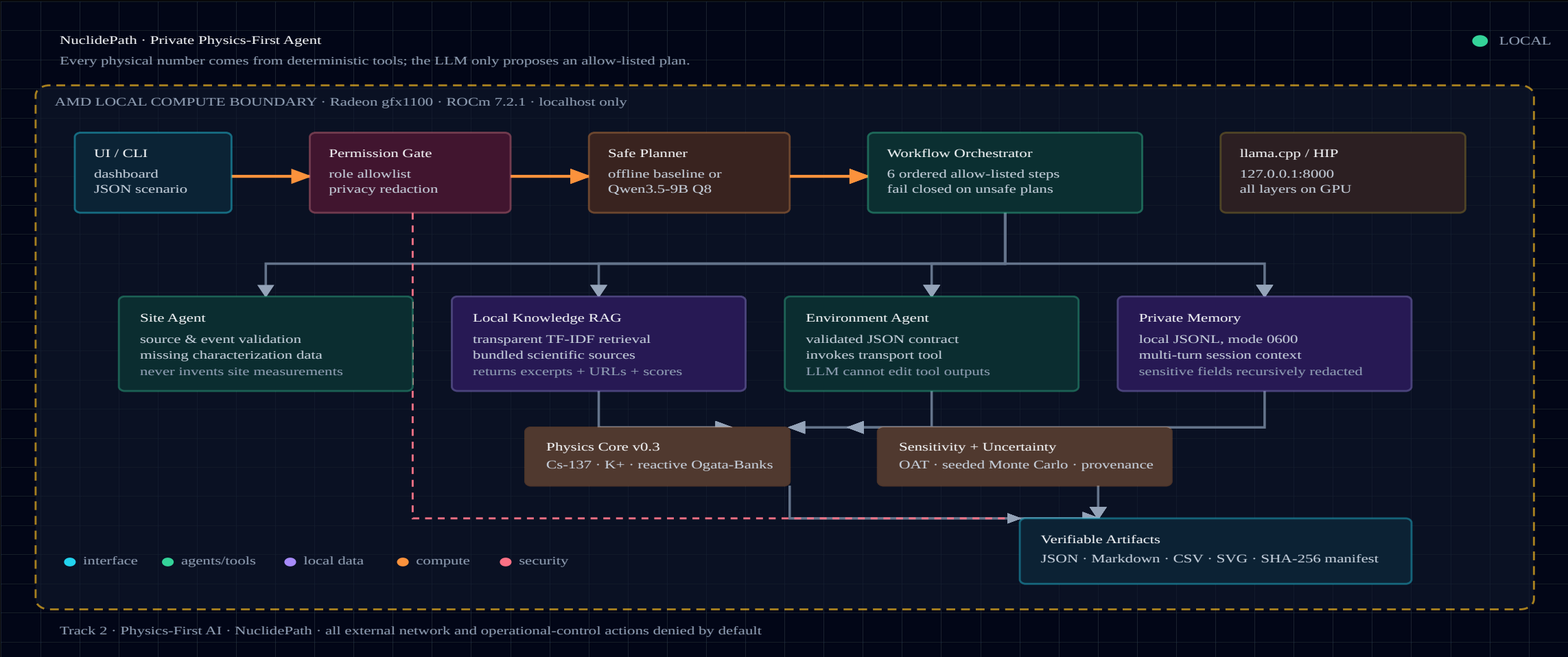
PLAN

TOOLS

EVIDENCE

A private workflow with an immutable physics boundary

The AMD-local model plans. Five deterministic subsystems execute and report.



All five private-agent capabilities are implemented

Each capability is exercised in the same end-to-end run and covered by tests.

01 Local retrieval

Ranked excerpts, local paths, source URLs and transparent scores.

VERIFIED

02 Tool invocation

Environment Agent calls a strict, versioned JSON physics contract.

VERIFIED

03 Multi-step planning

Deterministic fallback or Qwen3.5-9B returns six safe steps.

VERIFIED

04 Multi-turn memory

Session JSONL stays local, mode 0600, with explicit deletion.

VERIFIED

05 Permissions + privacy

Role allowlist, recursive redaction and denied operational actions.

VERIFIED

5/5

capability flags true

**One workflow. One
privacy boundary. No
cloud dependency.**

The LLM never calculates transport

Model 0.3 declares PDE, source and boundaries, then uses a verified analytical solution.

$$R \frac{\partial C}{\partial t} = D \frac{\partial^2 C}{\partial x^2} - v \frac{\partial C}{\partial x} - \lambda RC$$

$$C(x, 0) = 0 \quad \cdot \quad C(0, t) = C_0 \quad \cdot \quad C(\infty, t) = 0$$

$$A = \sqrt{(v^2 + 4\lambda RD)}$$

$$C/C_0 = \frac{1}{2} [\exp((v-A)x/2D) \operatorname{erfc}(z_-) + \exp((v+A)x/2D) \operatorname{erfc}(z_+)]$$

REACTIVE OGATA-BANKS

231

AMD core snapshot

0

LLM physics values

Independent analytical cases

Ogata-Banks limit · source boundary · reactive steady state · finite rejection

Explicit limitation

Constant-source 1-D screening — not a dose code, regulatory model or calibrated site prediction.

Multicomponent bridge (opt-in)

GCS K/Na competition + NH₄ on FES; extended K/Na/Ca/Mg path is process-qualified, not promoted.

Traceable post-merge science

The latest work adds deterministic, auditable layers around the canonical screening path.

388/2

AMD post-merge tests

4

ions · K/Na/Ca/Mg

2×

fixtures · brine/recharge

0

unqualified claims

PHREEQC bridge

Opt-in scenario compiler: declared water chemistry, CEC and Cs + K/Na/Ca/Mg into SOLUTION/EXCHANGE/TRANSPORT input

Arithmetic oracle

Independent machine-readable checks for units, grid mapping, non-negativity and exchange-site occupancy

Calibration gate

Dependency-free traceable Cs calibration/hold-out contract with fail-closed promotion; no data bundled

Benchmark registry

Separate EPA/Fuller/Dubus external Cs registry with source digests and calibration_eligible: false

PROCESS-QUALIFIED ONLY

Canonical transport-prototype-0.3 remains authoritative; the bridge is opt-in and does not replace it.

Qwen3.5-9B Q8 runs locally on Radeon + ROCm

The measured local model fits in VRAM and produced the verified allow-listed plan.

2,877.30

prompt tokens/s · 512 tokens

67.66

generation tokens/s · 128 tokens

19.9692 s

AMD v0.3 LLM pipeline

8.86 GiB

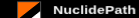
Q8 model size

GPU	AMD Radeon Graphics · gfx1100
Stack	ROCm 7.2.1 · llama.cpp HIP Release
Offload	-ngl 99 · all model layers
Context	8192 tokens · reasoning off
Network	127.0.0.1:8000 only
Integrity	SHA-256 verified model file

OFFLINE ≡ LLM-PLANNED

Identical physical runs
for the same scenario

PHYSICS IDENTICAL



physics-first private agent

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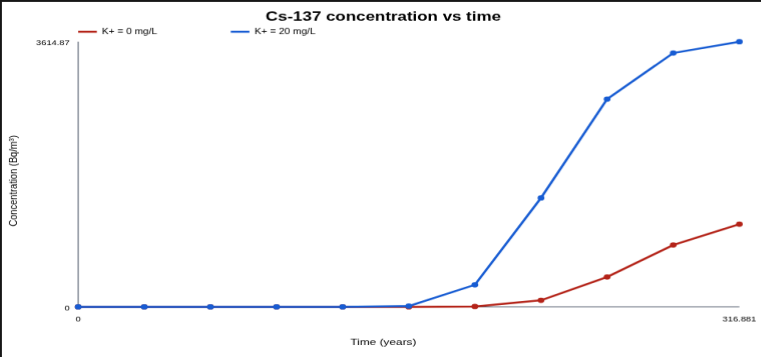
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Concentration over time - K⁺ comparison

Cs-137 concentration vs time



Time (years)	Concentration (Bq/m³) - K ⁺ = 0 mg/L	Concentration (Bq/m³) - K ⁺ = 20 mg/L
0	0	0
10	0	0
100	0	0
1000	0	0
10000	0	0
100000	0	0
1000000	~361.487	~3614.87

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Kd, K⁺, velocity, porosity, distance and C₀.

Six completed steps and 5/5 capability badges.

Ten downloadable artifacts plus a checksum manifest.

NuclidePath

physics-first private agent

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Seeded Monte Carlo intervals

Uncertainty intervals (P05–P95)

128 seeded Monte Carlo samples · median shown in amber · screening inputs only

Effective Kd (m²/kg)

P05 0.05162

P50 0.4303

P95 3.074

Retardation factor (–)

P05 250.3

P50 2179

P95 1.648e+04

Travel time (years)

P05 30.08

P50 724.2

P95 1.473e+04

Sampled maximum concentration (Bq/m³)

P05 0

P50 0.0002927

P95 4.995e+05

Final concentration (Bq/m³)

P05 0

P50 0.0002927

P95 4.995e+05

Quantile summary

Metric	P05	P50	P95
Effective Kd (m²/kg)	0.05162	0.4303	3.074
Final concentration (Bq/m³)	0	0.0002927	499,500
Retardation factor (–)	250.3	2,179	16,480
Sampled maximum concentration (Bq/m³)	0	0.0002927	499,500
Travel time (years)	30.08	724.2	14,730

Cs Kd: 0.05–5.0 m³/kg · IAEA TECDOC-2095

Hydraulics, dispersion, K+ and empirical competition are never presented as site truth.

Seeded Monte Carlo · P05/P50/P95 ·
JSON/CSV/SVG

WHY NUCLIDEPATH WINS

Private AI without scientific surrender.

LOCAL

Qwen3.5-9B + RAG + memory on AMD

VERIFIABLE

388 AMD post-merge tests + FP64 parity + SHA-256

USEFUL

One-click workflow from case to report

HONEST

PHREEQC bridge explicit; no unqualified science

THANK YOU

The proof

5/5

Track 2 capabilities

388

AMD tests (post-merge)

137.09x

FP64 platform

0

cloud calls

Track 2 · Physics-First AI · NuclidePath

Official AMD Track 2 PR package

TECHNICAL BUILD VERIFIED